

Alex Musick

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Summary

Software Engineering graduate with a focus on secure system design and full-stack development. Proficient in backend integration, cross-platform mobile, and agentic coding.

Experience

- **FYT Internship - 2025**
Implemented Stripe payment within a Firebase backend, optimizing payment flow and ensuring secure and compliant transaction handling.
- **Teton County Annex Security Analyst (Project-based) - 2024**
Evaluated network infrastructure across 50+ endpoints using nmap, identifying security gaps and drafting actionable mitigation plans for the IT team.

Education

Brigham Young University - Idaho

Bachelor of Science, Software Engineering - 2026

Relevant Coursework:

- CSE 310: Applied Programming
- CYBER 375: Security Analysis/Operations
- CYBER 370: Security Assessment
- CSE 212: Programming w/ Data Structures

Projects

- **Xbox Restore Combo Calculator:** Reverse-engineered an undocumented cryptographic function from C source code, and re-implemented it in Python.
<https://github.com/CitricDolphin1/xbox-restore-combo-calculator>
- **PesterMe:** Designed and developed a cross-platform scheduling assistant with Flutter.
<https://github.com/alex-musick/pesterme>
- **Deadline Dynamo:** Collaborated on a proof-of-concept hackathon project for a browser extension to intelligently sort school assignments in Canvas.
https://github.com/DanielGRasmussen/deadline_dynamo
- **Grapefruit:** Designed, implemented, and documented a semantic, object-oriented text adventure engine in Python.
<https://github.com/CitricDolphin1/grapefruit-tae>
- **Insaniquarium Deluxe Analysis:** Analyzed and documented the video game *Insaniquarium Deluxe* from the binary assembly in Ghidra.
<https://tcrf.net/Special:Contributions/CitricDolphin>

Proficiencies

- **Languages:** Python, Dart, Javascript, x86_64 Assembly, C, C++, HTML, CSS, Java, C#
- **Platforms:** Node.js, Godot, Oracle Cloud, Flutter, Firebase
- **Tools:** Wireshark, Security Onion, nmap, Burp Suite, Ghidra
- **AI Tools:** Claude Code, OpenCode, Llama.cpp